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Development of a Cost Analysis Spreadsheet for Calculating the Costs to Raise a Replacement Dairy Heifer¹

M. T. Gabler, P. R. Tozer, and A. J. Heinrichs

Department of Dairy and Animal Science
The Pennsylvania State University, University Park, PA 16802

ABSTRACT

Dairy operations have a variety of resources and objectives, such that the most economical method of obtaining replacement heifers is only determined by individual analysis of costs. The objective of this study was the development of a cost analysis spreadsheet and validation of that spreadsheet on milking and custom heifer operations throughout Pennsylvania. A cost analysis spreadsheet was developed with an Excel '97 Microsoft file. The spreadsheet estimated the costs to raise a replacement heifer by specific age classes for feed, labor, health, reproduction, bedding, facilities, equipment, mortality, and interest costs. The simplistic and broad-based nature of the spreadsheet was a key component in the spreadsheet's flexibility to estimate costs for a variety of operational objectives, feeding management, housing systems, and labor management.

A convenience sample of 16 milking operations and 14 custom heifer operations was evaluated to validate the cost analysis spreadsheet. Results from the validation are discussed to highlight the success and performance of the cost analysis spreadsheet. The average total cost to raise a replacement heifer for this data set was \$1124.06 and \$1019.20 for milking and custom heifer operations, respectively. Feed costs contributed 60.3 and 64.0% of the average total cost for milking and custom heifer operations, respectively. While no two operations are alike, individual operations possessing the ability to address costs to raise a replacement heifer can utilize critical information that can be used to improve operation profitability.

(**Key words:** heifer rearing, heifer economics)

INTRODUCTION

Acquiring healthy replacement heifers that calve between 22 and 24 mo of age represents a major expense to dairy operations. Because of the nature of replace-

ment heifer management, a dairy operation must invest feed, labor, and capital for 22 to 24 mo without receiving any realized benefits. Heinrichs (2) reports that heifer rearing represents the second largest expense on a dairy operation, approximately 20% of the total operation expenses, following only feed costs for the milking herd. Consequently, minimizing heifer rearing investments while maintaining the productive integrity of the replacement heifers should be the primary objective of replacement heifer management (4).

Research by Lin et al. (6) demonstrated the lifetime productive benefits of replacement heifers calving between 22 and 24 mo of age. Heifers calving at 23 mo acquired 107 more days of productive life, yielding 1475 more kg of milk, than replacement heifers calving at 26 mo. Although earlier age at calving has been proven beneficial (6), Clark et al. (1) determined that weight at calving influences first lactation milk yield four times more than age at calving alone. It has been shown that optimum calving weight of a first calf heifer is between 544 and 567 kg to maximize first lactation milk yield (5). For replacement heifers to calve between 22 and 24 mo, with an approximate BW of 544 to 567 kg, requires an increased rearing rate. While an increased rearing rate is required, research has shown that caution must be taken with heifer prepubertal growth rates (7, 8). Heifer prepubertal growth rates above 900 g/d results in a decrease in first lactation milk yield (7, 8).

The dynamics of heifer management make it difficult to meet one objective without negatively impacting another (4). Although the majority of dairy operations continue to raise their own replacement heifers, a growing number of dairy operations are contracting their heifers to specialized off-site heifer growers (3, 9).

Dairy operations have a variety of resources and objectives, such that the most economical method of obtaining replacement heifers is only determined by individual analysis of costs (9). The major objective of this study was the development of a cost analysis spreadsheet that is flexible to accommodate various operations, while accurately analyzing the costs to raise a replacement heifer. A secondary objective was validation of the cost spreadsheet on milking and custom heifer operations throughout Pennsylvania.

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Corresponding author: J. A. Heinrichs; e-mail: ajh@psuvm.psu.edu.

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MATERIALS AND METHODS

Development of a Cost Analysis Spreadsheet

A cost analysis spreadsheet was developed with an Excel '97 Microsoft file. Costs are calculated on a per head basis for feed, labor, health, reproduction, bedding, facilities, equipment, mortality, and interest with each of these aspects occupying an individual spreadsheet within the file (Appendix). All cost estimates are then transferred to a summary sheet that provides total and per day cost estimates to raise a replacement dairy heifer.

To provide flexibility for a variety of operations and specifics, cost estimates are also separated into age classes, which include 3 d of age until weaning, weaning until 6 mo of age, 6 mo of age until bred, and bred until prefresh. The cost estimates by age classes are transferred to a summary sheet that provides total cost and cost per day for the age classes. Reporting cost information in this manner allows the user to evaluate expenditures for different periods of rearing and realize opportunities for improvement.

Reproduction cost estimates include both artificial insemination and or the use of a service sire, which includes interest on purchase of a service sire, feed costs, and a yearly figure to cover all other service sire care related expenses. Facilities and equipment cost estimates incorporate a current value of either the facility or equipment acquired through a straight-line depreciation given the expected life of the facility or equipment. Facilities and equipment cost estimates also include the remaining life of the facility or equipment, number of days heifers are occupying the facility or using the equipment, interest, insurance, and tax on the ownership of the facility or equipment. Interest, insurance, and tax figures are acquired through the use of annual percentage rates. Mortality cost estimates takes in account the value of a replacement heifer, total cost to raise a replacement heifer, and the percent mortality for the operation at the specified age periods. Interest costs include the value of a replacement heifer, total cost to raise a replacement heifer, and an annual percent interest rate. A miscellaneous cost section is located in the summary sheet, where the operation enters a cost figure per head per age period to cover all inestimable costs such as but not limited to power, fuel for equipment, water, labor required to move animals, and transportation costs.

Validation of the Cost Analysis Spreadsheet

A convenience sample of 16 milking operations and 14 custom heifer operations were evaluated by using the cost analysis spreadsheet described above. These

Table 1. Life expectancy values placed on facilities and equipment.

Facilities	Expectancy
Permanent	30 yr
Nonpermanent ¹	15 yr
Equipment	
Automatic waterers	15 yr
Silo unloader	15 yr
Skid loaders	12 yr
Tractors	15 yr
Manure handling equipment	10 yr
Feed mixer wagons	15 yr
Operation vehicles	10 yr
Milk feeding equipment ²	3 yr

¹Nonpermanent facilities represent movable calf hutches and group housing.

²Milk feeding equipment represents milk bottles, buckets, and related necessities.

operations were evaluated throughout Pennsylvania between October 1998 and May 1999 to minimize possible seasonal variation in market values, which could alter the validity of the spreadsheet's calculated cost values. Replacement heifer cost data were collected by a single principal investigator and generally required 1.5 h per operation to complete. Replacement heifer cost data was entered into the cost analysis spreadsheet with results returned to the individual operations and pooled into a collective data set.

A month in the cost analysis spreadsheet comprised 30 d. Milking and custom heifer operation's prefresh date, the date at which costs associated with the replacement heifer enterprise was terminated in the validation study, was defined as 3 wk prior to expected first calving. A prefresh age defined as 3 wk prior to calving was implemented for equal comparison between operations, although the spreadsheet has the ability to calculate costs up until first calving. Labor hourly wage rates were based according to individual operations. All facilities and equipment were given life expectancies in which a straight-line depreciation was performed to estimate a current value (Table 1). If a facility or piece of equipment had outlived its life expectancy, an annual repair cost was assigned for the current value. The miscellaneous cost per age class was based according to the individual operation. The value of replacement heifers required to calculate mortality and interest costs, was dependent on the value of heifers at the individual operations. The annual interest rate was 4%, while the annual tax and insurance rates were 2%.

RESULTS AND DISCUSSION

The cost analysis spreadsheet was successful in that it allowed for a fair market value estimation of the

Table 2. Milking and custom heifer raising operation specifics

	Milking operations				Custom heifer operations			
	n	Avg	Min	Max	n	Avg	Min	Max
RHA ¹ , kg	16	10,038.18	8709.12	13,245.12	...	...	...	...
Number of heifers	16	141.69	40.00	420.00	14	111.14	32.00	250.00
Age at wean, d	16	57.94	30.00	120.00	8	53.50	42.00	70.00
Age at arrival ² , d	...	...	...	...	6	132.50	45.00	180.00
Age bred, mo	16	14.88	13.00	18.00	14	14.36	13.00	15.00
Age prefresh, mo	16	24.13	23.00	27.00	14	22.75	19.00	24.00

¹Rolling herd average.²Represents age the weaned heifer arrived to the custom heifer operation.

costs to raise heifers on various operations. Results from the operations analyzed are discussed to highlight the success of the cost analysis spreadsheet. Results are reported as total cost estimates and per day cost estimates. The simplistic nature of the spreadsheet was the key component in the spreadsheet's flexibility to estimate costs for a variety of operation's objectives, feeding management, housing arrangements, and labor management. All of the operations reported the information resulting from the spreadsheet allowed their operation to evaluate areas of strength and weakness within their heifer management. The second objective of the study was to validate the cost analysis spreadsheet on as wide a variation of operations as possible. This objective supported that economic decisions need to be based on individuals; not based on reported averages that did not allow for statistical comparison of the observed data.

The specifics of the operations analyzed are in Table 2. Of the 14 custom heifer operations analyzed, eight operations received heifers at or before 3 d of age, thus the cost to raise heifers from 3 d of age until weaning could be calculated. The remaining six custom heifer operations received heifers at an average age of 132.5

d. Data from these operations starts at 6 mo of age and continues until prefresh. Consequently there is a separation of the cost data for custom heifer operations, 3 d of age until prefresh and 6 mo of age until prefresh.

Milking operations had an average weaning age of 57.9 d while custom heifer operations had an average age at weaning of 53.5 d (Table 2). Age at weaning results in a savings of 8.4% for custom heifer operations over milking operations age at weaning for the farms in this data set (Figure 2). While total cost to raise a replacement heifer during the period 3 d until weaning is lowest as shown (Figure 1), this age class is by far the most expensive period for all farms, as demonstrated by the per day cost in Figure 2. The average total cost to raise a replacement heifer from weaning until 6 mo was \$170.75 and \$172.06 for milking and custom heifer operations, respectively (Table 3).

The average total cost to raise a replacement heifer from 6 mo until bred was \$313.27 and \$329.43 for milking and custom heifer operations, respectively (Table 3). Average age at breeding was slightly less for custom heifer operations (Table 2), yet the average total cost is larger. Total cost estimates for the age

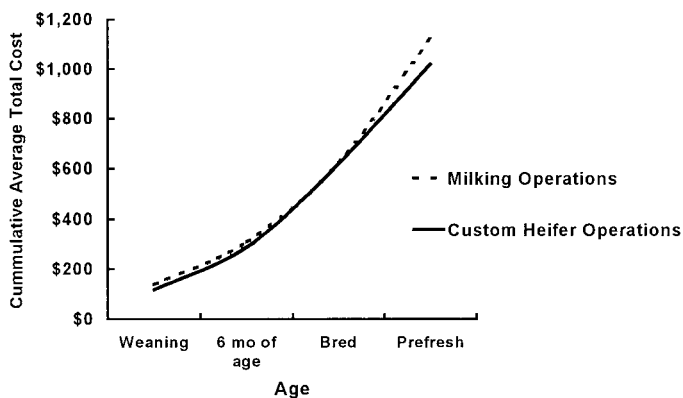
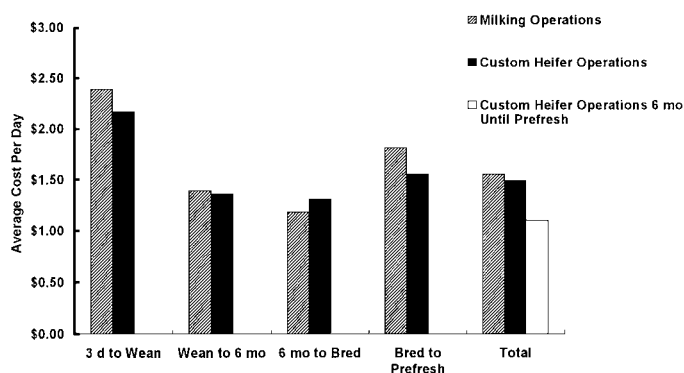
**Figure 1.** Cumulative average total cost from milking and custom heifer operations.**Figure 2.** Milking and custom heifer operation's average per day costs by age classes.

Table 3. Average total costs for specific age classes along with total cost to raise a replacement heifer for milking and custom heifer operations.

	Milking operations				Custom heifer operations			
	n	Avg	Min	Max	n	Avg	Min	Max
Total cost ¹	16	1124.06	896.86	1305.03	8	1019.20	798.18	1137.92
Total cost ²	16	815.04	608.15	1035.90	6	752.14	587.26	1096.37
3 d to Wean cost	16	138.27	63.46	299.02	8	115.97	70.70	165.55
Wean to 6 mo cost	16	170.75	68.34	270.60	8	172.06	114.36	217.44
6 mo to bred cost	16	313.27	117.11	426.32	14	329.43	99.79	559.49
Bred to prefresh cost	16	501.77	396.12	609.35	14	390.15	239.90	536.88

¹Total cost represents the total cost from 3 d of age until prefresh.

²Total cost represents the total cost from 6 mo of age until prefresh.

period 6 mo until bred for both milking and custom heifer operations vary widely, which may have attributed to this difference between lower age at bred and larger average total cost. Another likely possibility is the tendency for custom heifer operations to achieve more consistent growth by feeding a high nutrient ration during this period in hopes of decreasing age at first breeding, and ultimately decreased age at calving.

The age classes: weaning until 6 mo of age and 6 mo of age until bred are the periods of least expensive rearing as demonstrated by the average cost per head per day values (Figure 2). Average total cost to raise a replacement heifer for the age class bred until prefresh was \$501.77 and \$390.15 for milking and custom heifer operations, respectively (Table 3). The average age at prefresh for milking operations was 24.13 mo and 22.75 mo for custom heifer operations. Custom heifer operation's average age at prefresh represents a 17% savings over the average age at prefresh for milking operations in these farms.

Average total costs are broken into two categories due to the nature of the custom heifer operations. As shown by Figure 3, feed represents 60.3 and 64.0% of

the average total cost from 3 d of age until prefresh for milking and custom heifer operations, respectively. Feed represents 52.9% of the average total cost from 6 mo until prefresh for the remaining custom heifer operations. Labor was the second largest expense contributing 13.8 and 17.0% of the average total cost from 3 d until prefresh for milking and custom heifer operations, respectively. Of the custom heifer operations that raised heifers from 6 mo until prefresh, labor costs represented 13.4% of the average total costs.

Interest cost is an opportunity cost associated with the ownership, feed, labor, and capital required to raise a replacement heifer, compared with investing that capital in a bank. Ownership of the heifer largely influences interest expense. None of the custom heifer operations owned the heifers they raised, thus the average interest costs for these operations was lower than milking operations as shown in Figure 3.

Total cost for bedding, facilities, equipment, reproduction, health, and mortality are also presented in Figure 3. Mortality expenses for the operations in the study were very low due to the low average mortality rate for the entire rearing period; 0.95 and 0.76% for milking and custom heifer operations, respectively.

Figure 2 demonstrates the two most expensive periods of replacement heifer rearing based on per head per day costs. These two age classes are from 3 d until weaning and bred until prefresh. The findings presented by this study emphasize the importance of understanding the economics of raising replacement dairy heifers on an individual operation basis. Although no two operations are alike, possessing accurate economic figures such as those that can be provided by an analysis tool developed in this study can aid both milking and custom heifer operations improve operation profitability.

CONCLUSIONS

Specialization in the dairy industry and pressure to make sound economic and environmental decisions

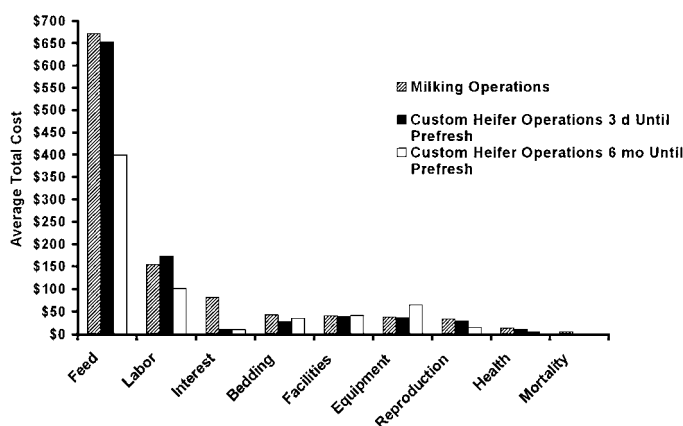


Figure 3. Milking and custom heifer operation's average total cost by task.

has created a need to continually evaluate each dairy management decision. A spreadsheet that evaluates the cost to raise a replacement heifer provides dairy operations the ability to make replacement heifer decisions based on accurate economic evaluation.

A wide range of variation exists in methods and management of replacement heifer raising. Knowing what it costs to raise a replacement heifer is essential when evaluating an operation's profitability and productivity. Results from the cost analysis spreadsheet developed in this study estimated the average total cost to raise a replacement heifer to be \$1124.06 for milking operations surveyed and \$1019.20 for custom heifer operations surveyed. Average per day cost was estimated to be \$1.55 and \$1.49 for milking operations and custom heifer operations, respectively. Feed cost contributed to over 60% of the total cost to raise a replacement heifer. This information should encourage industry to continue to acquire more economical methods and feed sources to supplement heifers. Selecting the most economic source or method of acquiring replacement heifers has major implications on dairy operation profitability but can be aided through the use of an accurate cost analysis spreadsheet and related information gained through this study.

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APPENDIX

Equations used to calculate costs per head per age class.

Variables

- z** = the inputs such as feed ingredients, labor tasks, synchronization agents, service sires, semen, heifers, bedding material, health agents, facilities, and equipment
- y** = the age class (3 d until weaning, weaning until 6 mo, 6 mo until bred, and bred until prefresh).
- A** = the amount of the input_z utilized by age class_y.
- D** = the number of days in which an input_z is utilized during age class_y.
- P** = the price per input_z utilized by age class_y.
- F** = the frequency of input_z utilized by age class_y.

$$\text{Feed cost} = (\mathbf{A}_{zy} \times \mathbf{D}_{zy})\mathbf{P}_{zy}$$

$$\text{Labor cost} = (\mathbf{T}_{zy}/\mathbf{NH}_{zy}) (\mathbf{W}_{zy}/60) (\mathbf{D}_{zy}/\mathbf{F}_{zy})$$

T = the time required to perform task_z during age class_y.

NH = the number of heifers in which task_z is performed during age class_y.

W = the wage per h for task_z during age class_y.

$$\begin{aligned} \text{Reproduction cost} = & \langle [(\mathbf{CD}_z \times \mathbf{ND}_z) + (\mathbf{P}_z \times \mathbf{SC}_z)] \\ & + (\mathbf{PV}_z - \mathbf{SV}_z) + \{(\mathbf{A}_z \times \mathbf{P}_z) + [(\mathbf{AZ}_z - \mathbf{AB}_z) \times 30]\alpha \\ & + \{(\mathbf{PV}_z - \mathbf{SV}_z) + (\mathbf{A}_z \times \mathbf{P}_z) + [(\mathbf{AC}_z - \mathbf{AB}_z) \times 30] \\ & \times .04\}/\mathbf{NY}_z \rangle + \mathbf{MS}_z \end{aligned}$$

CD = the cost per dose of semen_z.

ND = the number of doses of semen_z.

SC = the operation's services per conception on heifers for semen_z.

SV = the salvage value of service sire_z.

PV = the purchase value of service sire_z.

AC = the average age of heifers at confirmed pregnant in mo.

AB = the average age of heifers when first bred in mo.

NY = the number of heifers serviced by service sire_z per yr.

MS = any inestimable costs of having service sire_z, such as but not limited to bedding, health, labor to move, and manure handling.

$$\text{Bedding cost} = [(A_{zy}/NH_{zy}) \times (D_{zy}/F_{zy})]$$

$$\text{Health cost} = CA_{zy} \times NA_{zy}$$

CA = the cost per dose of vaccination_z administered to age class_y.

NA = the number of doses of vaccination_z administered to age class_y.

$$\begin{aligned} \text{Facility cost} = & [(CV_{zy} - SP_{xy}/NH_{zy})/L_{zy}] \\ & \times [(D_{zy}/30)12] + (R_{zy} + I_{zy} + IN_{zy} + TA_{zy}) \\ & \times [(CV_{zy} - SP_{xy}/NH_{zy})/L_{zy}] \times (D_{zy}/360) \end{aligned}$$

CV = the current single line depreciated value of facility_z in age class_y. If facility_z has outlived its life expectancy then an annual repair cost is entered.

SP = the value of the facility_x at the end of its life expectancy in age class_y.

L = the number of yr remaining of facility_z life expectancy in age class_y.

R = the annual percentage rate for repairs on facility_z in age class_y. If an annual repair cost has been entered for facility_z, then zero is entered.

I = the annual percentage rate for interest on facility_z in age class_y.

IN = the annual percentage rate for insurance on facility_z in age class_y.

TA = the annual percentage rate for taxes on facility_z in age class_y.

$$\begin{aligned} \text{Equipment cost} = & [(CV_{zy} - SP_{xy}/NH_{zy})/L_{zy}] \\ & \times [(D_{zy}/30)12] + (R_{zy} + I_{zy} + IN_{zy}) \\ & \times [(CV_{zy} - SP_{xy}/NH_{zy})/L_{zy}] \times (D_{zy}/360) \end{aligned}$$

$$\text{Mortality cost} = M_y \times [(D_y/30)/12] \times [(TC_y + VB_y + VE_y)/2]$$

M = the percent mortality rate for age class_y.

TC = the total cost to raise a heifer in age class_y.

VB = the value of a heifer at the beginning of age class_y.

VE = the value of a heifer at the end of age class_y.

$$\text{Interest cost} = I_y \times [(D_y/30)/12] \times [(TC_y + VB_y + VE_y)/2]$$